



SHOULDERS, BACK, AND CHEST

There are five major pairs of movements at the shoulder: (1) flexion and extension, (2) abduction and adduction, (3) external and internal rotation, (4) retraction and protraction, and (5) elevation and depression. The bones of the shoulder joint consist of the humerus (upper-arm bone), scapula (shoulder blade), and clavicle (collarbone). The scapula and clavicle essentially float on top of the rib cage. Therefore, a major function of many upper-back and chest muscles is to attach the scapula in the upper back and the clavicle in the upper chest to the rib cage and spine. This provides a stable platform for arm and shoulder movements. Of the five movement pairs, retraction and protraction and elevation and depression usually are classified as stabilization actions.

Most of the muscles involved in moving and stabilizing the shoulder bones are located posteriorly. The scapula is a much larger bone than the clavicle and has room for more muscles to attach. The posterior (back) muscles (figure 7.1) are the infraspinatus, latissimus dorsi, levator scapulae, rhomboids, subscapularis, supraspinatus, teres major, teres minor, and trapezius (attached to the upper posterior rib cage, vertebrae, and scapula), as well as the deltoid (figure 7.2) and triceps brachii (attached to the scapula and humerus; see chapter 6). The anterior (front) muscles (figure 7.3) are the pectoralis major (attached to the clavicle, anterior rib cage, and humerus), pectoralis minor, subclavius, serratus anterior (attached to the anterior rib cage and anterior scapula), biceps brachii, coracobrachialis, and deltoid (attached to the anterior scapula and humerus).

The shoulder, or glenohumeral, joint is a ball-and-socket joint formed by the head of the humerus and the glenoid fossa, a shallow scapular cavity that forms a socket for the humeral head. This joint is both the most freely moving joint of the body and the least stable. Upward movement of the humerus is prevented by the clavicle and the scapular acromion

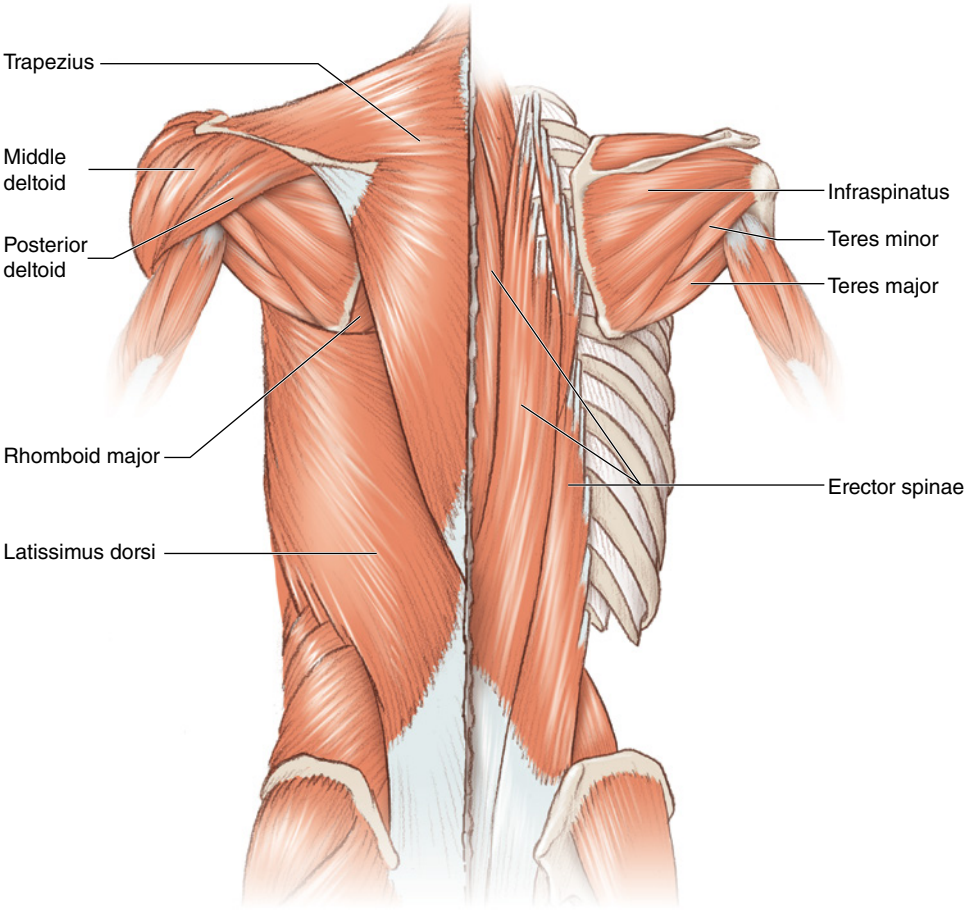


FIGURE 7.1 Back muscles.

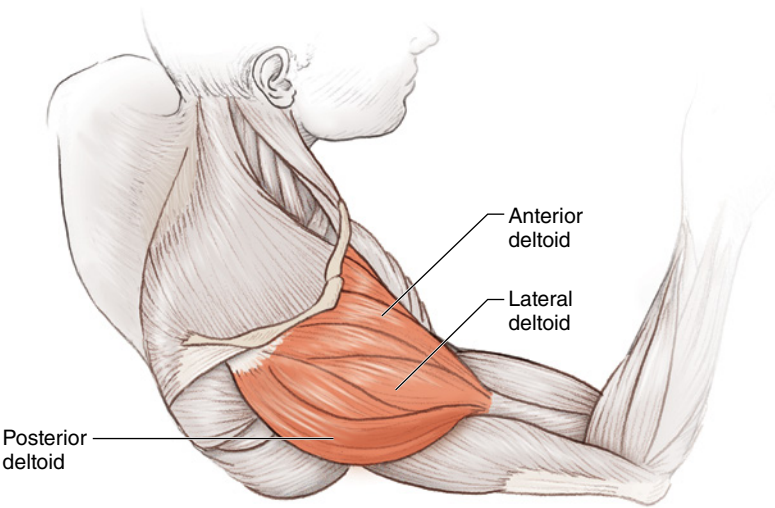


FIGURE 7.2 Deltoid muscle.

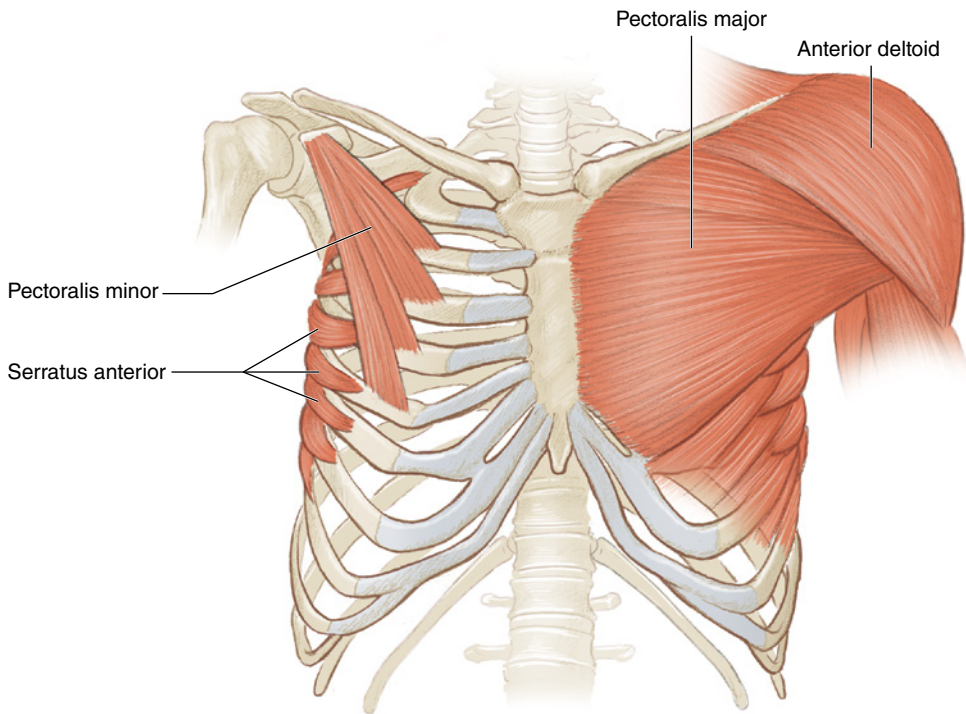


FIGURE 7.3 Chest muscles.

and coracoid processes, as well as by the glenohumeral ligaments and rotator cuff. Downward, forward, and backward humeral movements are limited by the humeral head's position in the glenoid labrum, a circular band of fibrocartilage that passes around the rim of the glenoid fossa to increase its concavity. Along with the glenoid labrum, the humerus is held in place by several ligaments and muscle tendons that together form the rotator cuff.

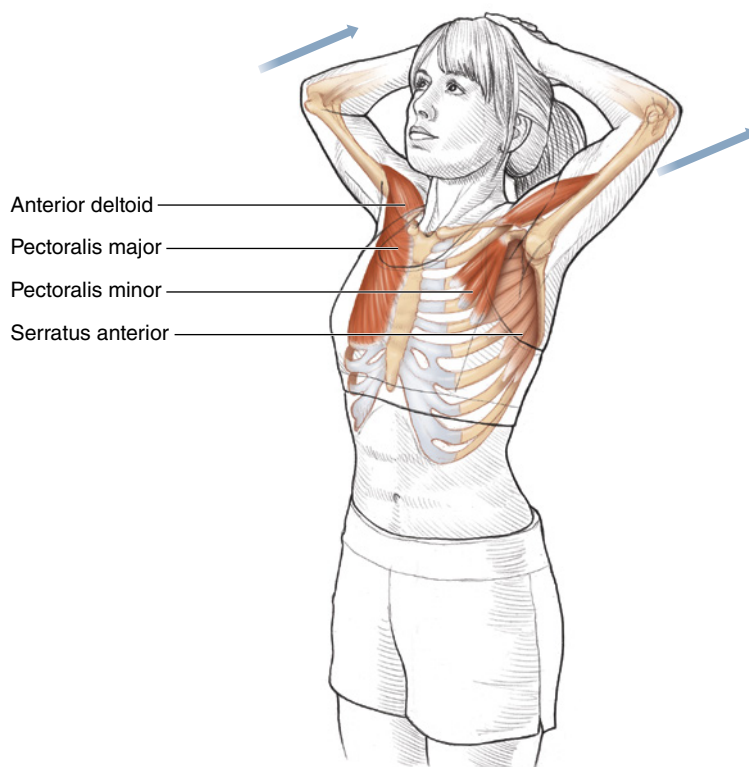
The whole humerus head and the glenoid fossa are surrounded by the joint capsule, a collection of ligaments. Major ligaments include the anterior and posterior sternoclavicular, costoclavicular, and interclavicular ligaments, which help connect the clavicle to the rib cage. The coracohumeral, glenohumeral, coracoclavicular, acromioclavicular, and coracoacromial ligaments help interconnect the humerus, scapula, and clavicle bones. The major muscles and tendons providing rotator cuff stability are the infraspinatus, subscapularis, supraspinatus, and teres minor. Because these muscles attach more superiorly (atop the shoulder), most dislocations occur inferiorly (downward from the shoulder).

Because the shoulder muscles are a major component of shoulder stability, shoulder flexibility—the amount of possible movement in a particular direction—in all five movement pairs (e.g., extension and flexion) is greatly controlled by both the strength of the muscles and the extensibility of the antagonist muscles involved in the movement. Shoulder abduction, the range of motion away from the midline of the body, is limited by the flexibility of the ligaments in both the shoulder and the joint capsule and by the humerus hitting the acromion and the superior rim of the glenoid fossa (or shoulder impingement). Shoulder adduction, the range of motion toward the midline of the body, is additionally limited by the arm meeting the trunk. Shoulder flexion range of motion is limited by the tightness of both the coracohumeral ligament and the inferior portion of the joint capsule. Coracohumeral ligament flexibility influences shoulder extension range of motion along with shoulder impingement. Shoulder internal rotation is restricted by the flexibility of the capsular ligaments, while external rotation range of movement is limited by rigidity of the coracohumeral ligament and the tightness of the superior portion of the capsular ligaments. Additional factors for elevation include the tension of the costoclavicular ligament along with the joint capsule. For depression, the other restrictors are the interclavicular and sternoclavicular ligaments. Finally, protraction is limited by tightness in both the anterior sternoclavicular and posterior costoclavicular ligaments, while retraction is limited by tightness in both the posterior sternoclavicular and anterior costoclavicular ligaments.

It is important to maintain proper balance between strength and flexibility in all shoulder muscles. Common complaints associated with the musculature of the shoulders, back, and chest are tight muscles and muscle spasms in the neck (middle and upper trapezius), shoulder (trapezius, deltoid, supraspinatus), and upper back (rhomboids and levator scapulae). Interestingly, the tightness felt in these muscles is usually a result of initial tightness in their antagonist muscles. In other words, tight muscles in the upper chest caused the tightness felt in the upper back. Tight chest muscles (e.g., the pectoralis major) cause a constant low-level stretch on the muscles of the upper back. Eventually, this low-level stretch elongates the ligaments and tendons associated with the upper-back muscles. Once these ligaments and tendons become elongated, the tone in their associated muscles falls dramatically. To reclaim the lost tone, the muscles must increase their force of contraction. Increased force in turn causes more stretch of the ligaments and tendons, and increased muscle contraction must compensate for that. Hence, a vicious cycle commences.

The best way to prevent or stop this cycle is to stretch the anterior shoulder and chest muscles. As the flexibility of these muscles increases, the tightness of the posterior muscles is reduced. Immediately after stretching, the strength of the muscles is diminished. It is a good idea to stretch the opposing muscles just before and immediately after working any group of muscles. If this is done three or more times a week, the muscles will actually increase in flexibility and gain strength. Stretching will also reduce the frequency of tightness for any group of muscles. Furthermore, shoulder impingement can occur with improper balance between shoulder muscle strength and flexibility. Because the gap between the humerus and scapular process is narrow, anything that further narrows this space, such as tight muscles, can result in impingement, leading to pain, weakness, and loss of movement.

BEGINNER SHOULDER FLEXOR STRETCH



Execution

1. Stand upright and interlock your fingers.
2. Place your hands on top of your head.
3. Contract your back muscles, and pull your elbows back toward each other.

Muscles Stretched

Most-stretched muscles: Pectoralis major, pectoralis minor, anterior deltoid

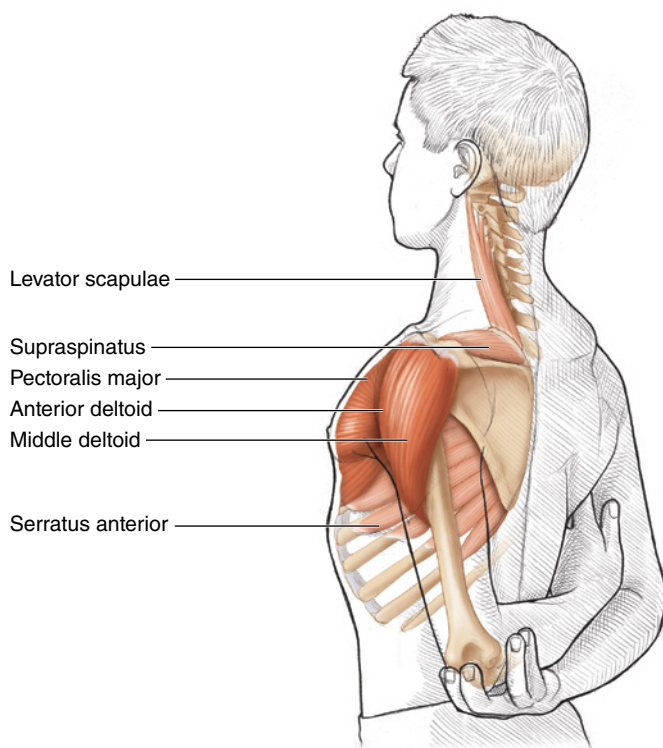
Less-stretched muscle: Serratus anterior

Stretch Notes

Poor posture is the primary reason for tight shoulder flexor muscles. Poor posture is commonly seen when a person hunches forward or works with the arms extended out in front. Tightness usually is accompanied by tight neck extensors. Having both groups of muscles tight increases the chances of developing a “vulture neck,” in which the head position looks like the protruding head of a vulture, and contributes to breathing problems. Injuries, either acute or overuse, that lead to shoulder impingement, shoulder bursitis, rotator cuff tendinitis, or frozen shoulder can also lead to tight shoulder flexors.

When any of these conditions are severe, it is difficult to stretch the flexors without pain. This stretching activity places a low stretch stress on the musculature and hence is easy to tolerate. When you feel less stretch when doing this activity, it is best to advance to one of the other shoulder flexor stretching activities.

INTERMEDIATE SHOULDER FLEXOR STRETCH



Execution

1. Sit upright on a backless chair or stand, with the left arm behind the back and the elbow bent to about 90 degrees.
2. Place feet shoulder-width apart, with the toes pointing forward.
3. Grasp the left elbow, forearm, or wrist, depending on your flexibility, with the right hand.
4. Pull the upper left arm across the back and up toward the right shoulder.
5. Repeat this stretch for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left pectoralis major, left anterior deltoid, middle deltoid

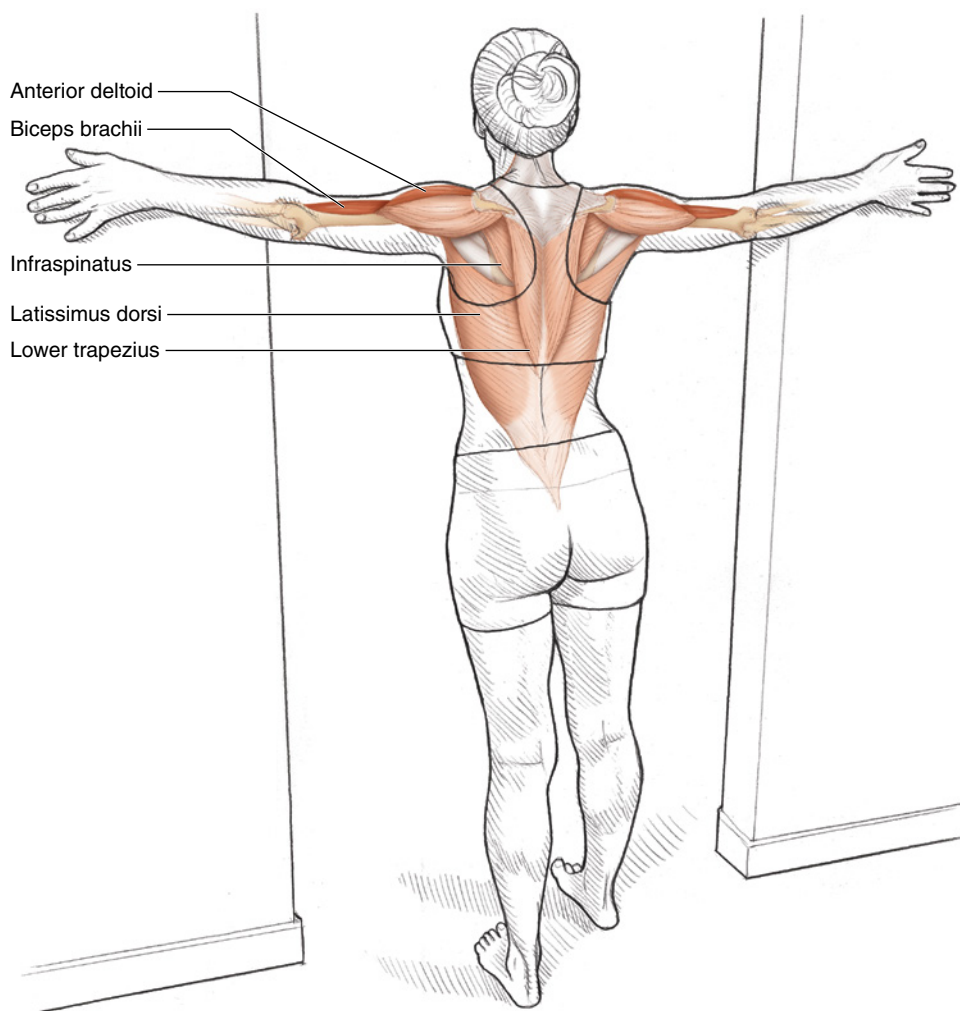
Less-stretched muscles: Left levator scapulae, left pectoralis minor, left supraspinatus, left serratus anterior, left coracobrachialis

Stretch Notes

This stretch is excellent for overcoming a vulture neck or rounded, hunched shoulders arising from poor posture. It also helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder. This exercise provides a better stretch than the beginner shoulder flexor stretch, but it is best to use this stretch only after you have progressed through the beginner exercise and find it difficult to apply any stretch at the beginner level.

If you cannot reach the elbow, then grasp the wrist. When pulling on the wrist, it is easy to pull the arm across the back, but remember that the best effect comes from pulling upward as well as across. Also, keep the elbow locked at a near 90-degree angle. Changing the alignment of the back will also influence the magnitude of the stretch. If you cannot keep the back straight, arching the back is preferable to bending at the waist. Just be careful; it is easy to lose balance when doing this stretch while both arching the back and standing. If maintaining balance while standing is a problem, do this stretch while sitting on a stool or chair.

ADVANCED SHOULDER FLEXOR STRETCH



Execution

1. Stand upright while facing a doorway or corner.
2. Place the feet shoulder-width apart, with one foot slightly in front of the other.
3. With straight arms, raise your arms to shoulder level, and place the palms on the walls or doorframe with the thumbs on top.
4. Lean the entire body forward.

Muscles Stretched

Most-stretched muscles: Pectoralis major, anterior deltoid, coracobrachialis, biceps brachii

Less-stretched muscles: Infraspinatus, latissimus dorsi, subclavius, lower trapezius

Stretch Notes

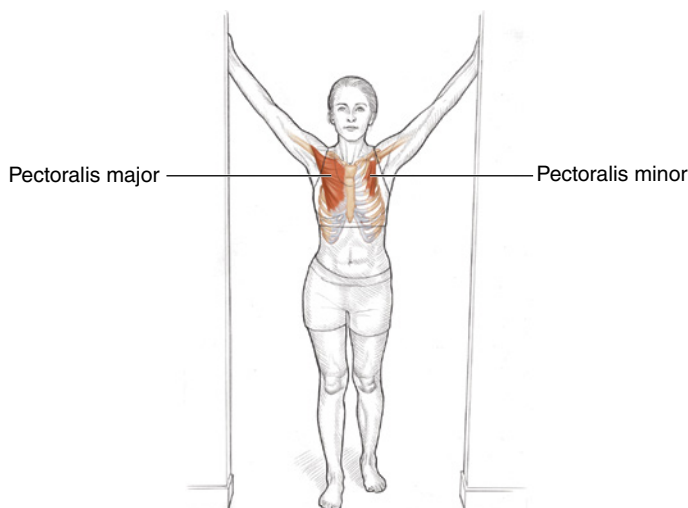
This stretch is excellent for overcoming a vulture neck or rounded, hunched shoulders arising from poor posture. It also helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder. However, if you have any of the aforementioned problems, it is better to start with the beginner stretch and work your way up to the advanced stretch. This exercise provides a better stretch than either the beginner or intermediate shoulder flexor stretches, and it is better to use if you can tolerate the pain or discomfort it may produce.

To get the maximum benefit during the stretch, keep the elbows locked and the spine straight. The greater the forward lean, the better the stretch. Forward lean is controlled by how far the lead foot is in front of the chest at the starting position. Hence, place the foot forward only enough to maintain balance. It is possible to do the neck extensor stretch simultaneously with the shoulder flexor stretch, but without the hands pushing down on the head. However, without the hands pushing down on the head, the neck extensor stretch will be of a lower intensity than if it were done by itself.

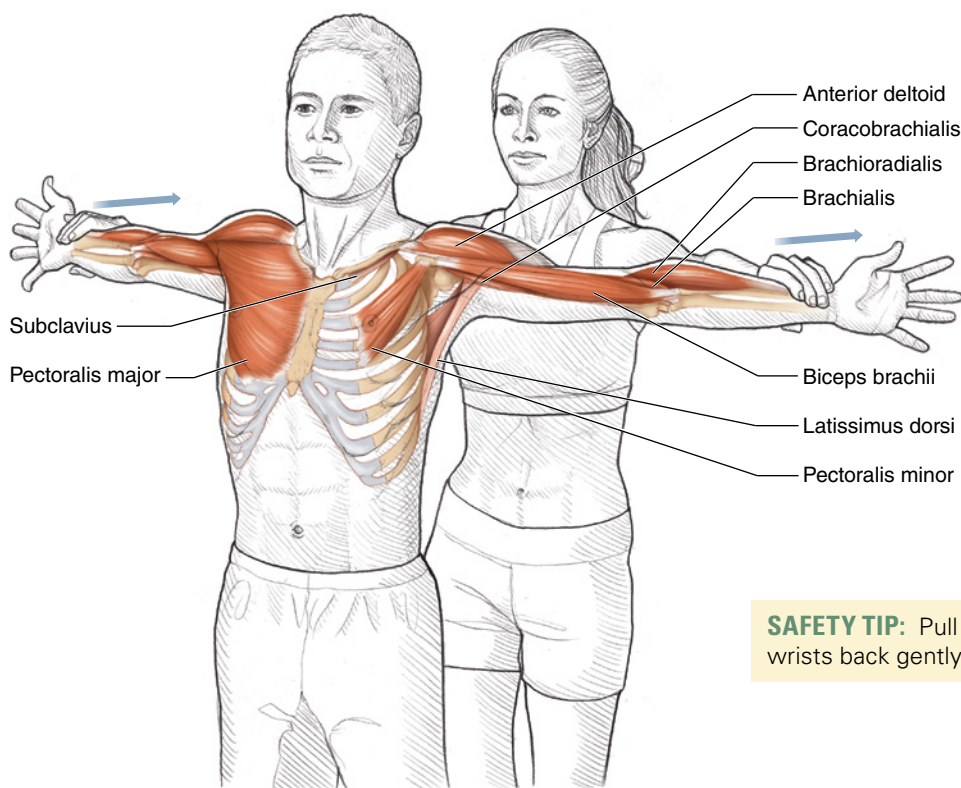
VARIATION

Shoulder Flexor and Depressor Stretch

By elevating the arms above parallel, you can include the pectoralis minor as one of the major muscles being stretched. Stand upright while facing a doorway or corner, with the feet shoulder-width apart and one foot slightly in front of the other. Keeping the arms straight, raise the arms high above the head, and place the palms on the walls or doorframe. Lean the entire body forward.



ASSISTED SHOULDER AND ELBOW FLEXOR STRETCH



SAFETY TIP: Pull the wrists back gently.

Execution

1. Stand upright or sit on the floor for more stability.
2. If standing, place feet shoulder-width apart with one foot slightly in front of the other. If sitting, sit on the ground with both legs extended in front of you.
3. Extend both arms parallel to the floor.
4. Point the hands slightly back.
5. A partner stands behind you facing your back and grabs hold of each arm at the wrist.
6. The partner pulls the wrists toward each other while being careful not to overstretch the joint.

Muscles Stretched

Most-stretched muscles: Pectoralis major, pectoralis minor, anterior deltoid, coracobrachialis, biceps brachii, brachialis, brachioradialis

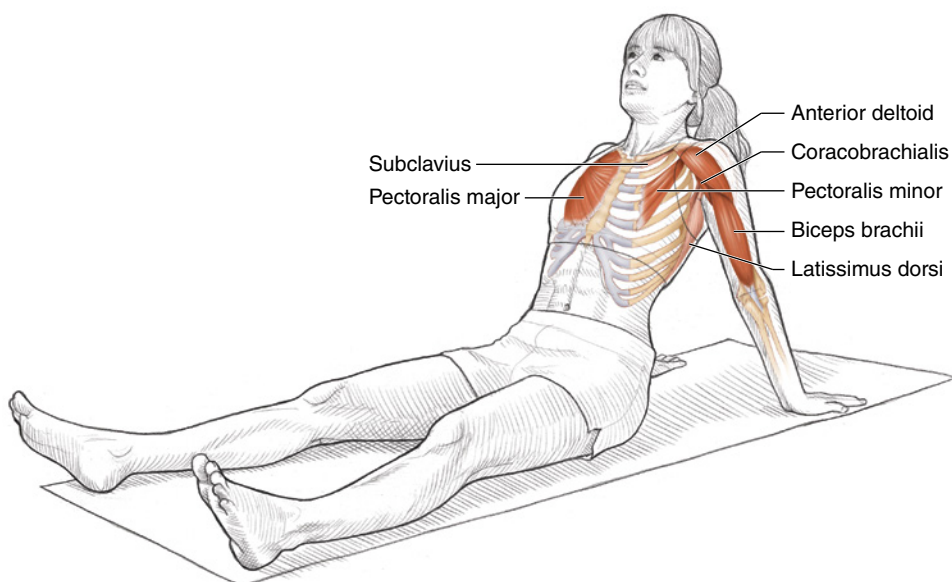
Less-stretched muscles: Latissimus dorsi, lower trapezius, subclavius

Stretch Notes

This stretch is excellent for overcoming a vulture neck or rounded, hunched shoulders arising from poor posture. It also helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder. Additionally, this stretch helps prevent what many people call muscle boundness, or rounded and forward-thrusted shoulders combined with an inability to completely straighten the arms. This stretching activity is one of the better exercises for both the shoulder and elbow flexors. The partner can modify the stretch to tailor it to beginner through advanced by simply stretching to the point of pain toleration.

It is important that the partner assisting with this stretch not be overly aggressive when pulling the wrists together. An overly aggressive stretch can result in muscle strains and, in extreme cases, shoulder dislocation. Moreover, as the wrists get closer to each other, people have a tendency to lean back to reduce the pain. If you find yourself leaning back, it is a good idea to bend at the waist and lean slightly forward at the start of the stretch.

SEATED SHOULDER FLEXOR, DEPRESSOR, AND RETRACTOR STRETCH



Execution

1. Sit on the floor with the legs straight.
2. While keeping the arms straight, place the palms on the floor, fingers pointed back, about 1 foot (30 cm) behind the hips.
3. While keeping the arms straight, lean back toward the floor.

Muscles Stretched

Most-stretched muscles: Pectoralis major, anterior deltoid, coracobrachialis, biceps brachii, pectoralis minor

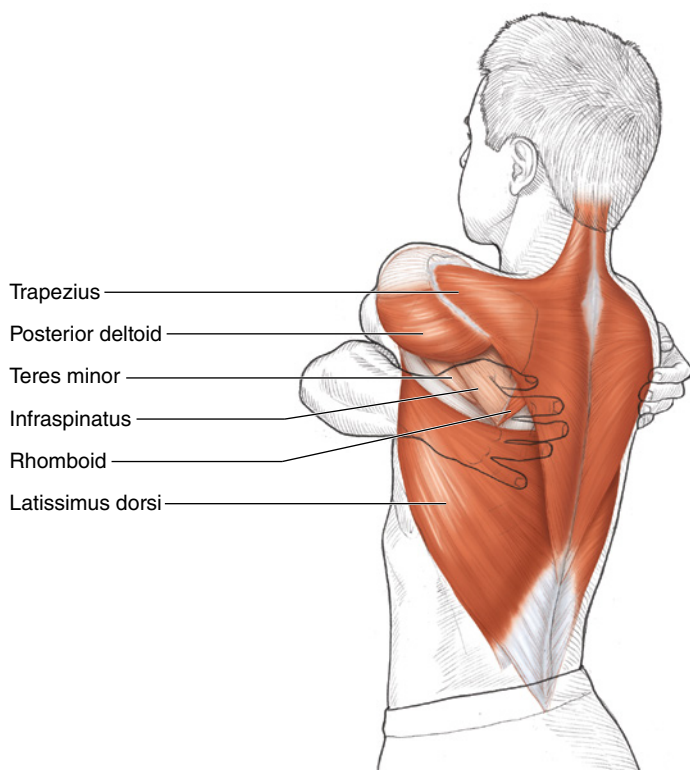
Less-stretched muscles: Latissimus dorsi, lower trapezius, subclavius, rhomboids

Stretch Notes

This stretching activity is one of the better unassisted exercises for stretching both the shoulder and elbow flexors simultaneously. It is an excellent stretch for overcoming a vulture neck or rounded, hunched shoulders arising from poor posture. It also helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder. Additionally, this stretch helps prevent what many people call muscle boundness, or rounded and forward-thrusted shoulders combined with an inability to completely straighten the arms.

To maximize the stretch, keep the arms straight. If it is difficult to refrain from bending the arms, place the hands closer to the hips. Moving the hands farther from the hips can increase the stretch. To keep the body from sliding along the floor, you may need to brace the soles of the feet against a wall. Sitting on a mat with the hands on a hard surface will increase the stretch as well as add comfort.

BEGINNER SHOULDER EXTENSOR, ADDUCTOR, AND RETRACTOR STRETCH



Execution

1. Stand upright with feet shoulder-width apart, toes pointing straight forward.
2. Wrap your arms around your shoulders as if you were hugging yourself, placing the arm on top that is the most comfortable.
3. Pull your shoulders forward.

Muscles Stretched

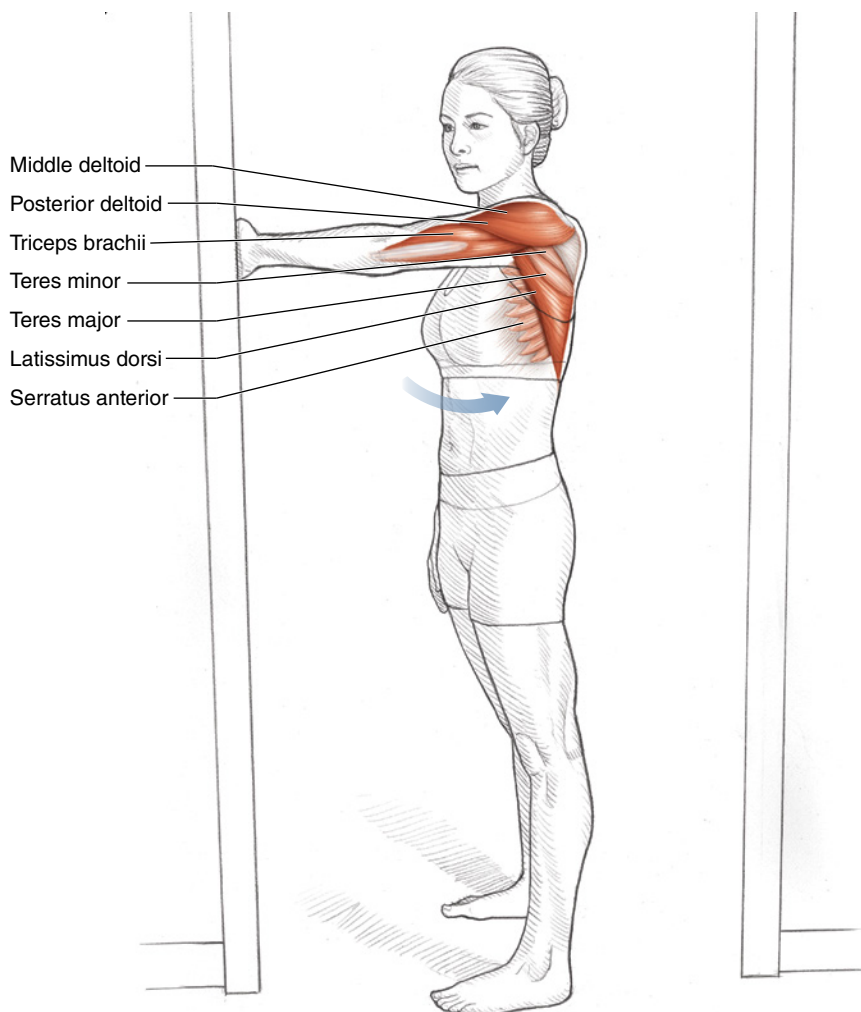
Most-stretched muscles: Posterior deltoid, latissimus dorsi, trapezius, rhomboids

Less-stretched muscles: Teres minor, infraspinatus

Stretch Notes

Poor posture overworks the deltoids, lats, traps, and rhomboids, causing tightness. This stretch relieves many of the aches and pains felt between the shoulder blades. Conversely, these muscles can also become tight from disuse or by doing limited activities with the arms below shoulder level. Tightness in these muscles makes overhead work, such as painting a ceiling, washing overhead windows, or doing a dumbbell overhead press, harder and more painful. This stretching activity places a low stretch on the musculature and is the best one to start with if you have extremely tight muscles. Also, this stretch helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder.

INTERMEDIATE SHOULDER EXTENSOR, ADDUCTOR, AND RETRACTOR STRETCH



Execution

1. Stand upright inside a doorway while facing a doorjamb, with the doorjamb in line with the right shoulder.
2. Place the feet shoulder-width apart, with the toes pointing straight forward.
3. Bring the left arm across the body toward the right shoulder.
4. Pointing the thumb down, grab hold of the doorjamb at shoulder level.
5. Rotate the trunk in until you feel a stretch in the posterior left shoulder.
6. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left posterior deltoid, left middle deltoid, left latissimus dorsi, left triceps brachii, left middle trapezius, left rhomboids

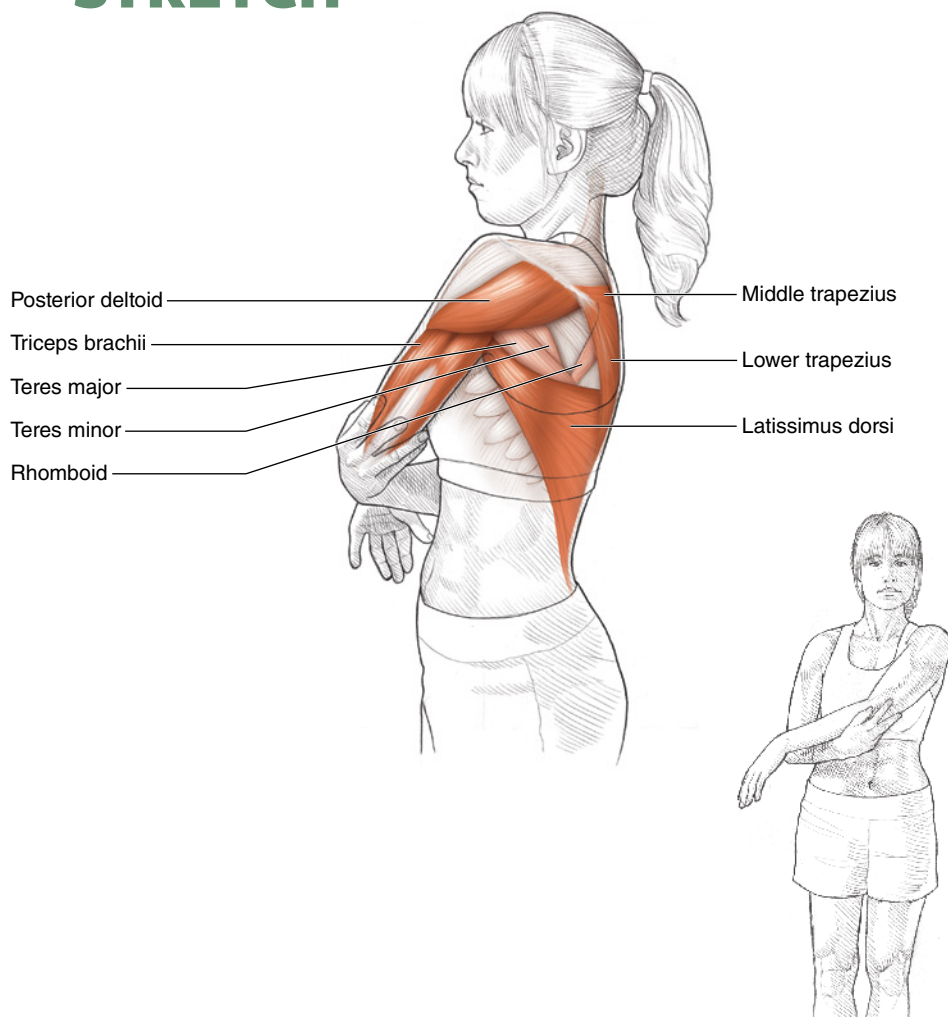
Less-stretched muscles: Left teres major, left teres minor, left supraspinatus, left serratus anterior

Stretch Notes

Poor posture overworks the deltoids, lats, triceps, traps, and rhomboids, causing tightness. This intermediate stretch places more stretch on these muscles. It relieves many of the aches and pains felt between the shoulder blades better than the beginner stretch. Conversely, these muscles can also become tight from disuse or by doing limited activities with the arms below shoulder level. Tightness in these muscles makes overhead work harder and more painful. This stretching activity places a greater stretch on the musculature than the basic shoulder extensor, adductor, and retractor stretch. Also, this stretch helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder.

To get the maximum benefit of this stretch, keep the elbow locked. Over time, as the muscles become more flexible, to keep the elbow locked you will need to grasp the doorframe above the level of the shoulder. Raising the hand does not diminish the major benefits of this stretch. However, as the hand gets higher above shoulder level, the stretch on the rhomboids decreases while the stretch on the serratus anterior increases.

SHOULDER ADDUCTOR, PROTRACTOR, AND ELEVATOR STRETCH



Execution

1. Stand upright with the feet shoulder-width apart.
2. Bring the left arm across the front of the body, with the left hand near the right hip.
3. With the right hand, grab the left elbow.
4. With the right hand, try to pull the left elbow down and around the right side of the body.
5. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left posterior deltoid, left latissimus dorsi, left triceps brachii, left lower middle trapezius

Less-stretched muscles: Left teres major, left teres minor, left supraspinatus, left levator scapulae, left rhomboids

Stretch Notes

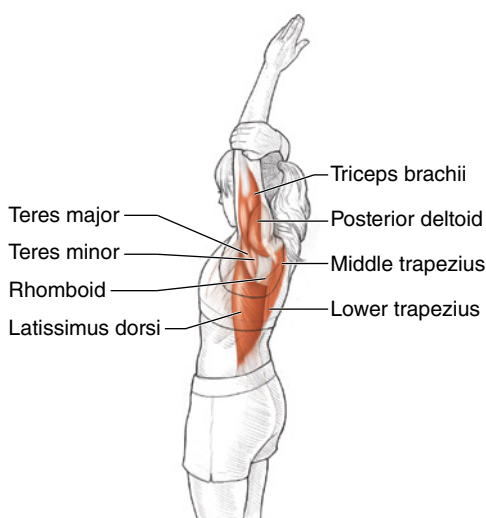
Tightness in the deltoids, lats, triceps, and traps makes overhead work harder and more painful. Thus, this stretch makes it easier to perform throwing actions as well as around-the-house activities such as painting and window cleaning. Also, this stretch can help relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder.

To maximize the stretch, do not raise the shoulder or bend at the waist. If it is not possible to bring the hand toward the hip, try to come as close as possible. As long as the arm is below the shoulders, the stretch will be effective.

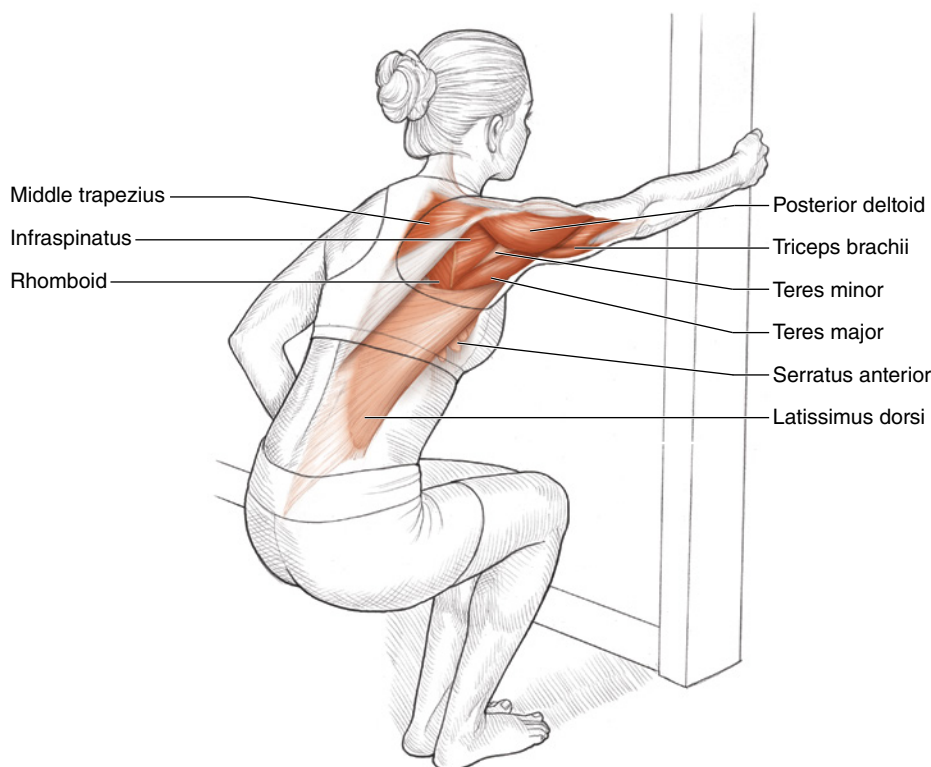
VARIATION

Overhead Shoulder Adductor, Protractor, and Elevator Stretch

Bringing the arm above the shoulder places more stretch on the elevators and protractors and is more beneficial for high overhead activities. Stand upright with the feet shoulder-width apart. Raise the left hand high above the head, and bring the left arm up against the left side of the head. Then, with the right hand, grab hold of the left elbow and try to pull the left elbow behind the head, past the left ear. Repeat these steps for the opposite arm.



SHOULDER ADDUCTOR AND EXTENSOR STRETCH



Execution

1. Squat while facing a doorway, with the right shoulder lined up with the left side of the doorjamb.
2. Extend the right arm through the doorway. Grab the inside of the doorjamb at shoulder level with the right hand.
3. While keeping the right arm straight and the feet firmly planted, lower the buttocks toward the floor.
4. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Right posterior deltoid, right middle trapezius, right triceps brachii, right teres major, right rhomboids, right infraspinatus

Less-stretched muscles: Right latissimus dorsi, right teres minor, right supraspinatus, right serratus anterior

Stretch Notes

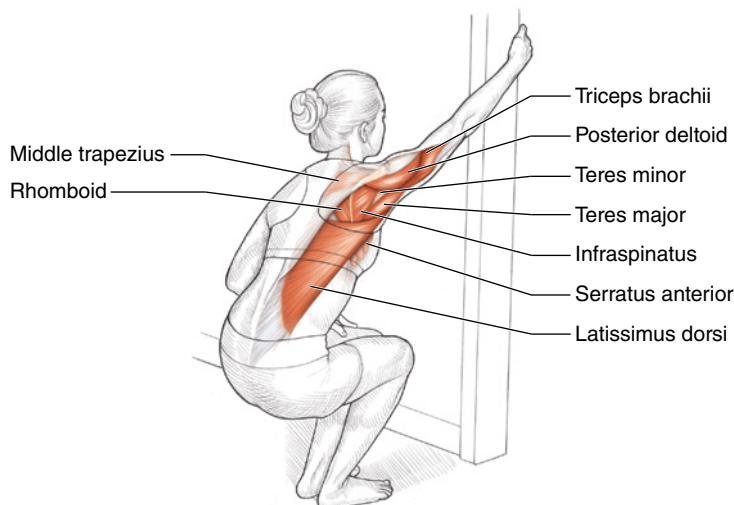
Although poor posture has a negative effect on both sides of the body and results in overall tightness, most people use one arm more than the other, so the muscles on one side can become tighter from disuse. This is especially possible when doing overhead work such as painting, window washing, or overhead presses. These activities may become harder and more painful. Thus sometimes you may need to stretch one side more than the other. Because this stretch mimics single-arm overhead work, it is better suited for problems arising from one side being tighter than the other. Also, by stretching one side singularly with gravity assistance, this stretch allows for a greater amount of stretch than any of the other stretches that work similar muscles. Moreover, this stretch relieves many of the aches and pains felt between the shoulder blades.

A lower squat yields a greater stretch, but it increases the pressure and strain on the knee joints. Therefore, do not squat so low that you feel pain in the legs or knees. To reduce strain on the knees, change the point where you grab the doorjamb. Changing the position of the grasp, however, influences the amount of stretch placed on the various muscles (see variation). Regardless of where you grasp, keep the back straight or arched. Do not bend forward at the waist. To get an even greater stretch, inwardly rotate the trunk.

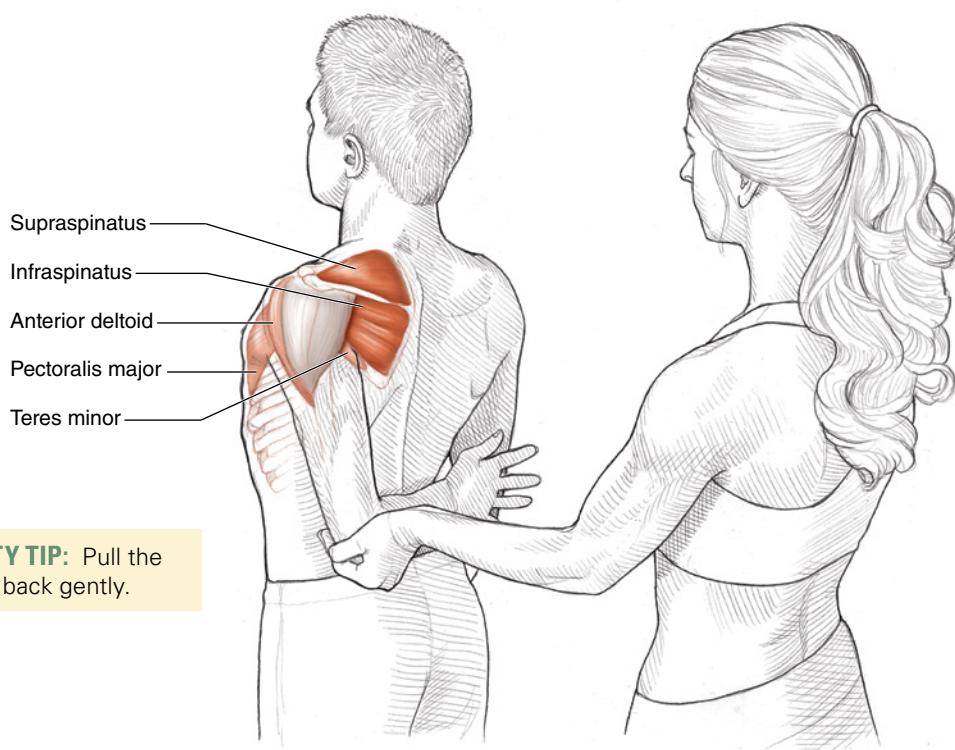
VARIATION

Overhead Shoulder Adductor and Extensor Stretch

Grasping the inside of the doorjamb above head level reduces the stretch on the middle trapezius and allows a greater stretch of the posterior deltoid, latissimus dorsi, triceps brachii, teres major, and infraspinatus. Begin the stretch by squatting in front of a doorway, with the right shoulder in line with the left side of the doorway. Extend the right arm through the doorway, and, with the right hand, grab the inside of the doorjamb several inches above your head. Increase the stretch by lowering the buttocks toward the floor. Repeat again for the opposite side.



ASSISTED SHOULDER ABDUCTOR STRETCH



SAFETY TIP: Pull the elbow back gently.

Execution

1. Stand upright with the feet shoulder-width apart and the toes pointing straight forward.
2. Bring your left arm behind your back, with the elbow bent to 90 degrees.
3. A partner stands behind you facing your back and grasps the left elbow.
4. The partner gently pulls the elbow back and up toward the head, taking care not to pull suddenly or with great force.
5. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left supraspinatus, left infraspinatus

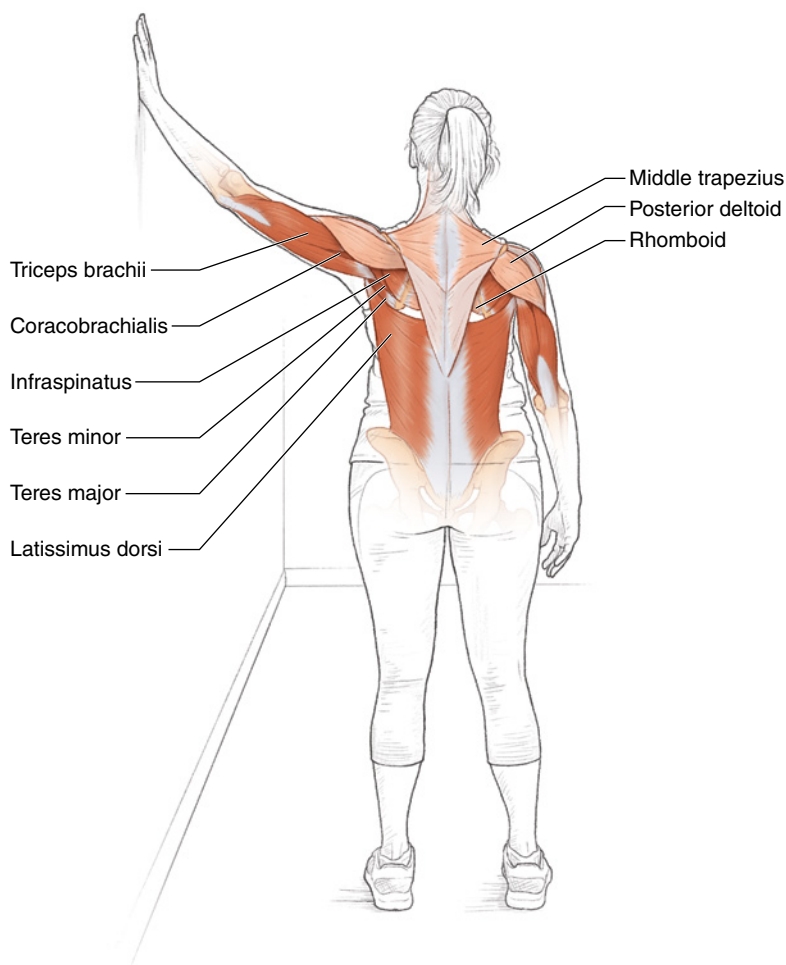
Less-stretched muscles: Left anterior deltoid, left pectoralis major, left teres minor, left coracobrachialis

Stretch Notes

The supraspinatus and infraspinatus muscles can become tight when a person does either repeated forward pushing actions, as when using a walk-behind lawn mower, or downward pulling actions, such as raising something off the ground using a block-and-tackle pulley system. The supraspinatus especially is always working during overhead movements and so can be easily strained when it fatigues. This stretch can also help relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder.

If someone has twisted your arm behind your back, you know that this movement can be painful. The pain is magnified if these muscles are very tight. Therefore, the person assisting with this stretch needs to proceed slowly when pulling the arm up and back.

SHOULDER ADDUCTOR STRETCH



Execution

1. Stand upright with the feet shoulder-width apart and the left side of the body facing a wall about an arm's length away.
2. Place the palm of the left hand on the wall at shoulder height.
3. Gradually “walk” the fingers up the wall as high as you can without pain.
4. Flatten the hand against the wall, and lean the weight on the hand.
5. Hold this position for the desired length of time.
6. Repeat these steps for the opposite side.

Muscles Stretched

Most-stretched muscles: Left coracobrachialis, left infraspinatus, left latissimus dorsi, left teres major, left teres minor, left triceps brachii

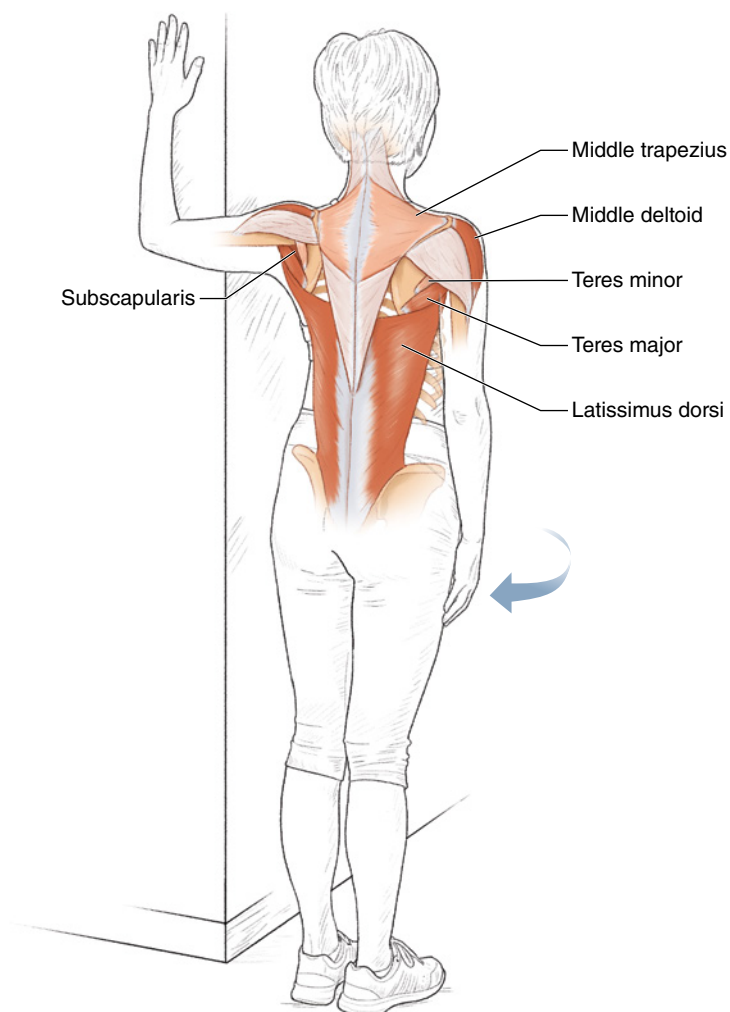
Less-stretched muscles: Left posterior deltoid, left middle trapezius, left rhomboids

Stretch Notes

This stretch makes it easier to perform throwing actions as well as overhead around-the-house activities such as painting and window cleaning. Also, this stretch can help relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, and frozen shoulder. This activity especially helps a frozen shoulder. Keeping the weight resting on the fingers as they “walk” up the wall will help strengthen the shoulder abductors as well as stretch the shoulder adductors when you rest the weight at the terminal height.

Keeping the arm in a straight line between the body and the wall places the stretch equally on the anterior and posterior muscles. Moving the hand toward the front of the body places more stretch on the posterior muscles and moving the hand toward the back stretches the anterior muscles more. You can combine this stretch with the intermediate lower-trunk lateral flexor stretch in chapter 5.

SHOULDER INTERNAL ROTATOR STRETCH



Execution

1. Stand upright while facing an outside corner, with the left shoulder in line with the corner's edge.
2. Place the feet shoulder-width apart and the toes pointing toward the wall.
3. Place the chest as close as possible to the wall, and flex the left elbow to 90 degrees.
4. With the elbow at 90 degrees, place the forearm and palm flat against the other wall of the outside corner.
5. Keeping the forearm and palm flat against the wall, step backward, rotating the chest away from the wall.
6. Rotate until the stretch is felt in the left shoulder and back.
7. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left anterior deltoid, left middle deltoid, left latissimus dorsi, left pectoralis major, left subscapularis, left teres major

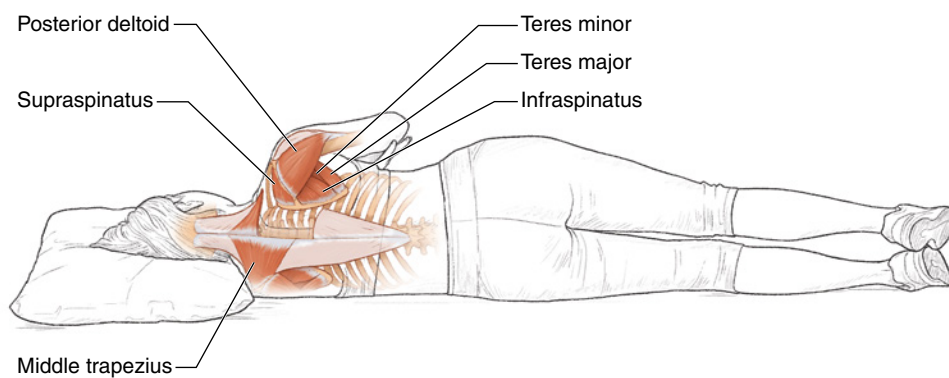
Less-stretched muscles: Left pectoralis minor, left teres minor, left middle trapezius

Stretch Notes

Frozen shoulder, or adhesive capsulitis, develops from decreased use of the arm, especially when you are recovering from a medical condition or procedure. Symptoms are characterized by stiffness and pain in the shoulder, typically beginning gradually, but worsening over time. While you need care from skilled medical services if you have a frozen shoulder, this exercise can help prevent and mitigate this condition. Also, this stretch helps relieve the pain associated with shoulder impingement, shoulder bursitis, rotator cuff tendinitis, upper crossed syndrome, and tight pectoral muscles caused by poor posture.

To get the maximum benefit of this stretch, keep the elbow locked at 90 degrees. Over time, as the muscles become more flexible, you can try different elbow angles more and less than 90 degrees to determine whether isolated muscles need additional stretch. Once you can rotate the body past 45 degrees from the wall, you can increase the stretch by leaning forward. If an outside corner is difficult to find, or if you have trouble keeping the elbow locked, you can grasp either the inside of a doorframe or the inside edge of a door. Grasping the doorframe or door does not diminish the major benefits of this stretch.

SHOULDER EXTERNAL ROTATOR STRETCH



Execution

1. Lie on the left side with the left arm positioned away from the body at least 10 degrees below horizontal.
2. Tilt the back posterior with the bodyweight resting on the lateral scapula border.
3. Bend the left elbow to 90 degrees and point the fingers skyward.
4. Grab the left elbow with the right hand and rest the right forearm on the left forearm.
5. Gently use the right forearm to press the left forearm down.
6. Repeat these steps for the opposite arm.

Muscles Stretched

Most-stretched muscles: Left infraspinatus, left posterior deltoid, left teres minor

Less-stretched muscles: Left middle trapezius, left supraspinatus, left teres major

Stretch Notes

In addition to the muscles being stretched, this exercise is one of the best for stretching the posterior capsule of the shoulder. It is important that the scapula remain stationary during this stretch. The scapula is moving if you can easily press the forearm completely flat while feeling very little stretch in the shoulder or back. To help ensure the scapula stays stationary, support the head so that it is perpendicular to the shoulder, and support the back so that it stays tilted posteriorly. To place the body in the proper position, start by lying supine (faceup) with the arms out in a T position. Roll to one side and bring the outstretched arms together until the palms meet. Now, tilt backward.

It is important to keep the arm-shoulder angle less than horizontal (90 degrees). Doing this stretch with the joint at 90 degrees can cause shoulder impingement and can damage the posterior capsule. Grabbing the elbow with the opposite hand and using forearm-to-forearm pressure helps reduce shoulder impingement. Never use one hand to push the other hand down.

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